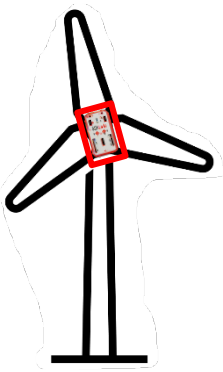
Experiment 4: Rolling rigid body.



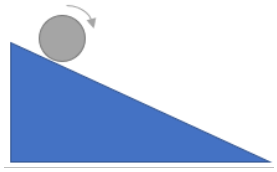
The Engineering Company '*WindmillsRUs*' designs new windmills. During testing, they would like to monitor a windmill's behaviour by measuring the wings' angular velocity under different operating conditions. Because building a new windmill is expensive, they always test on scale models. In practice, they attach an iOLab to the wings, and the iOLab rotates with the wings (see illustration).

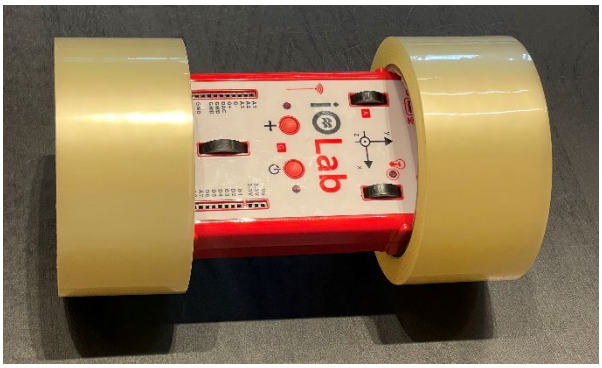
Unlike the actual windmill, the iOLab is not very small compared to the scale windmill. This means the iOLab contributes to the moment of inertia of the model's wings. Therefore, the engineers need to correct their measurements by accounting for this additional moment of inertia. In this experiment, you will determine the moment of inertia of the iOLab to be able to correct its effect on the scale system. This makes it possible to relate the scaled model to the actual windmill.

<u>The exercise:</u>

You should plan and execute several attempts to determine the moment of inertia of a rigid body. The rigid body under investigation is the iOLab with two tape rolls attached; however, you should determine the moment of inertia of the iOLab about the y-axis.

You should make two independent determinations of the moment of inertia, one based on an energy consideration, and one based on the rotational analogue of Newton's second law.





(OBS: Do not press the tape rolls past the small notches on the iOLab.)

Write a short lab report about the experiment.

Reflection:

Write three to four lines about each of the questions below. Include these on an extra page 4.

1) How did you approach today's experiment: did you have expectations of the final result, or did you only get the result after you had the data?
2) What did you learn from today's experiment?
3) What do you think you were supposed to learn from today's experiment?